
The Probe Budget of Stochastic Curvature Supervision: A Variance Frontier and Residual-Adaptive Thinning for Hessian-Trained Interatomic Potentials

Working paper

Abstract

Supervising machine-learning interatomic potentials with second derivatives through stochastic Hessian–vector products has made curvature training affordable: one random probe per molecule per minibatch yields an unbiased projected-Hessian loss at near-force cost. But one probe per molecule is a convention, not a measured operating point, and the probe budget is the estimator’s true resource. We ask how few probes curvature supervision needs at fixed statistical quality, and answer with a theory–measurement pair. Thinning probes across a minibatch with exact Horvitz–Thompson compensation preserves unbiasedness for any adapted allocation; its gradient-variance cost admits a closed-form lower bound governed by two checkpoint-measurable coefficients, the heterogeneity of per-molecule gradient magnitudes and the probe-noise share. Measured on the reactive-chemistry benchmark and reference protocol of Projected Hessian Learning, these coefficients are stationary across training and imply a hard feasibility frontier: at a $2\times$ gradient-variance cap, no allocation in this estimator class—including one with oracle knowledge of every gradient magnitude—can thin probes five-fold, while two-to-three-fold thinning is available. A practical rule (residual-adaptive probe thinning, RAPT) prices each molecule’s probe by the square root of its own cached residual with a size discount and an annealed budget, adds no model evaluations, and approaches the oracle frontier. Across five paired seeds at the reference protocol, RAPT runs at $2.13\times$ fewer probes per minibatch step (time-averaged) with audited per-dataset variance ratios within the cap and downstream energy, force, and Hessian accuracy statistically indistinguishable from dense probing on all nine dataset–property cells; a pre-registered beyond-frontier variant at $3.0\times$ violates the cap as predicted and exhibits small but statistically resolved Hessian-cell degradation, bracketing the cap empirically. The frontier coefficients can be measured on checkpoints of any Hessian-supervised run before committing to a budget, and the analysis predicts the feasible reduction grows with system size—where curvature supervision is most expensive.

1 Introduction

Machine-learning interatomic potentials (MLIPs) reproduce quantum-chemical energies $E(\mathbf{R})$ and forces $\mathbf{F} = -\nabla E$; supervising *second* derivatives—the Hessian $H_{(i\alpha),(j\beta)} = \partial^2 E / \partial r_i^\alpha \partial r_j^\beta$ —improves curvature-sensitive predictions: vibrational frequencies, transition-state geometry, robustness far from equilibrium [3, 5, 6, 7, 8, 9, 10]. The obstacle is cost: $(3N)^2$ Hessian entries per molecule multiply an energy–force pipeline’s epoch cost by an order of magnitude. Projected Hessian Learning (PHL) [1] removes most of this by replacing the explicit Hessian loss with an unbiased Hutchinson-type estimate from one Hessian–vector product (HVP) per molecule per minibatch: a probe v with $\mathbb{E}[vv^\top] = I$ turns the Frobenius error $\|\tilde{H} - H\|_F^2 / (3N)^2$ into the projected

$\|\tilde{H}v - Hv\|^2/(3N)^2$, one extra double-backward pass, matching full-Hessian training at $\sim 4\%$ of its cost [1].

One probe per molecule per minibatch is, however, a convention, not a measured operating point: each minibatch of B molecules pays B HVPs, and nothing in the estimator requires this. The probe budget is the natural resource measure of stochastic curvature supervision—a property of the estimator, not the hardware, and the dominant step cost beyond benchmark scale. We ask: *how few probes does curvature supervision actually need, at fixed statistical quality?* A reduction must preserve three invariants of dense probing: exact unbiasedness of loss and gradient, variance within $2\times$, and statistically indistinguishable downstream accuracy.

We answer in three parts. **(1) A thinned estimator with exact compensation.** Curvature is supervised on a random subset of each minibatch: molecule m is probed with an inclusion probability π_m computed before the step, its projected loss reweighted by $1/\pi_m$ (Horvitz–Thompson); unbiasedness of loss and gradient is exact for any adapted π (Proposition 1). **(2) A measurable variance frontier.** A bias-free probe cut is not free: for *every* allocation rule—including one with oracle knowledge of per-molecule gradient magnitudes—the variance ratio to dense probing obeys $R \geq 1 + (h/\rho - 1) \bar{s}^2/\bar{\sigma}_v^2$, with ρ the probe fraction, $h \in (0, 1]$ the heterogeneity of per-molecule gradient magnitudes, and $\bar{\sigma}_v^2/\bar{s}^2$ the probe-noise share (Proposition 2); all are measurable on checkpoints, and inverting the bound gives the smallest admissible probe fraction at a cap: $\rho^* = h/(1 + (\bar{R} - 1)\bar{\sigma}_v^2/\bar{s}^2)$. Measured on the PHL benchmark, the frontier is far from the one-probe convention but also far from zero. **(3) A practical rule that approaches it.** *Residual-Adaptive Probe Thinning* (RAPT) sets inclusion probabilities proportional to $\hat{\ell}_m^{1/2} N_m^\beta - \hat{\ell}_m$ an exponential moving average of the molecule’s own past projected residuals, a free byproduct of training, N_m its size—under a deterministic budget schedule $\rho(e)$ that spends probes early, where the frontier is measured hostile, and thins once the residual landscape stabilizes. RAPT adds no HVPs, nothing beyond an $O(B)$ waterfilling, and no change to probe law, loss, architecture, or optimizer.

On the OpenREACT-CHON-EFH benchmark [2], with the reference ANI-style architecture and protocol of [1], RAPT delivers $2.13\times$ fewer probes per step (time-averaged 188 of 400) at matched schedule, floor-phase variance ratios 1.70–1.87 (95% CIs below the $2\times$ cap) on Test, IRC, and NMS, and energy/force/Hessian RMSEs statistically indistinguishable from dense probing on all nine (property $\times$ dataset) cells across five paired seeds ($p \geq 0.17$). Equally important, the frontier explains what a larger reduction would require: more heterogeneous per-molecule gradient magnitudes than this dataset exhibits, or estimator structure outside the independent per-molecule probing class (we map the escape routes and measure why the natural ones fail here).

The negative space is part of the content: stochastic-trace folklore suggests extreme probe sparsity should be tolerable, but at minibatch scale probe noise and limited cross-molecule heterogeneity cap admissible thinning well above zero. Larger claims must give up variance parity or unbiasedness, or import new estimator structure.

2 Background and related work

Hessian supervision for MLIPs. Force supervision is standard across architectures [11, 12, 13, 14, 15]; second-order supervision is an active frontier: full-Hessian training improves accuracy and stability at $\sim 25\times$ epoch cost on our benchmark [3, 1], and direct Hessian prediction [8], transition-state Hessian learning [5], distillation via random Hessian rows [6], phonon fine-tuning [7], and Hessian datasets at scale [9, 10] attest to curvature labels’ value. PHL [1] is our reference: it supervises $\|(\tilde{H} - H)v\|^2/(3N)^2$ with fresh Gaussian probes, one per molecule per minibatch, inside $\mathcal{L} = \mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H$, $(\lambda_F, \lambda_H) = (0.30, 0.09)$, and reaches full-Hessian accuracy in the randomized-probe regime. We hold all of this fixed, varying only *how many* probes are drawn and *where they go*.

Stochastic trace estimation. The Hutchinson estimator [4] underlies matrix-free trace computation; modern variants cut probe counts by orders of magnitude for a *fixed* matrix by spending budget on a low-rank sketch acting as a control variate (Hutch++ [17], XTrace [18], dynamic variants [19]; see also [20]). These do not transfer here: the “matrix” is a per-molecule residual $A_m = B_m^\top B_m$ that changes every optimizer step and is touched once per epoch, so no budget exists to amortize a sketch, and the target is a *gradient* of the trace in the network parameters, not the trace. What transfers is the

accounting: probes as the resource, minimal budget at a variance target, our frontier playing the role of that literature’s lower bounds.

Importance sampling and subsampling in SGD. Sampling examples by loss or gradient-norm estimates is well developed [21, 22, 23, 24, 25, 26], as is unbiased sparsification with $1/p$ compensation [27]. Our setting differs in three ways. First, no example is skipped: every molecule keeps its energy and force terms; only the auxiliary curvature channel is thinned, so the estimator is a partially-subsampled multi-task loss, not example selection. Second, the stochasticity includes a per-item probe draw whose noise persists at $\pi = 1$; this probe-noise floor changes the optimal allocation exponent (magnitude^{1/2}, not magnitude) and dominates the frontier. Third, the importance signal is a cached byproduct of the estimator itself, so allocation adds zero forward or backward passes—the regime loss-based selection targets [24, 25], here with exact compensation. Uniform thinning (the same-budget analogue of naive subsampling) is compared in all ablations.

Curvature supervision at reduced label cost. Row/column-subsampled Hessian supervision [6, 10] reduces the *label* side; PHL’s one-column baseline is its in-framework analogue [1]. Probe-count reduction is orthogonal—thinning how often the estimator is evaluated, not what it projects—and composes with any probe law. We know of no prior work treating the per-minibatch probe budget of Hessian-supervised training as the optimization variable, stating the variance-parity constraint, or reporting the feasibility frontier.

3 Thinned stochastic curvature supervision

3.1 Setting and notation

We inherit the PHL objective [1]. A minibatch $\mathcal{B}_t = \{1, \dots, B\}$ contributes

$$\mathcal{L}_t = \frac{1}{B} \sum_{m \in \mathcal{B}_t} [e_m(\theta) + \lambda_F c_m(\theta)] + \lambda_H \hat{\mathcal{L}}_{Ht}, \quad e_m = \frac{(\tilde{E}_m - E_m)^2}{\sqrt{N_m}}, \quad c_m = \frac{\|\tilde{\mathbf{F}}_m - \mathbf{F}_m\|^2}{3N_m}, \quad (1)$$

with $\lambda_F = 0.30$, $\lambda_H = 0.09$. Dense probing (PHL) uses

$$\hat{\mathcal{L}}_{Ht}^{\text{dense}} = \frac{1}{B} \sum_{m \in \mathcal{B}_t} \ell_m(v_m), \quad \ell_m(v) = \frac{\|\tilde{H}_m v - H_m v\|^2}{(3N_m)^2}, \quad v_m \sim \mathcal{N}(0, I_{3N_m}) \text{ i.i.d.}, \quad (2)$$

where $\tilde{H}_m v$ is one batched HVP (a double backward through the force graph) and $H_m v$ is a matrix–vector product against the stored reference Hessian. Because $\mathbb{E}_v[vv^\top] = I$,

$$\mathbb{E}_v[\ell_m(v)] = L_m(\theta) := \frac{\|\tilde{H}_m - H_m\|_F^2}{(3N_m)^2}, \quad (3)$$

the exact per-molecule Hessian loss [4, 1]. A step’s *probe count* is the number of molecules whose model-side HVP is evaluated; dense probing costs B per step; we report full-run time averages. ∇ is the gradient in the parameters θ .

3.2 The thinned estimator and its exactness

Let $\mathcal{F}_{t-1}$ be everything realized before step t (parameters, past probes and selections). RAPT draws $b \in \{0, 1\}^B$ with independent $b_m \sim \text{Bernoulli}(\pi_m)$, the $\pi \in [\pi_{\min}, 1]^B$ being $\mathcal{F}_{t-1}$ -measurable (cached statistics; §3.4), and replaces (2) by the Horvitz–Thompson-compensated sum

$$\hat{\mathcal{L}}_{Ht}^{\text{thin}} = \frac{1}{B} \sum_{m \in \mathcal{B}_t} \frac{b_m}{\pi_m} \ell_m(v_m), \quad v_m \sim \mathcal{N}(0, I_{3N_m}) \text{ fresh, only for } b_m = 1. \quad (4)$$

The step’s probe count is $\sum_m b_m$, expectation $\rho_t B$ set by the budget schedule. Molecules with $b_m = 0$ keep their energy and force terms in (1); only the curvature term is omitted, and the $1/\pi_m$ weight compensates exactly:

Proposition 1 (Unbiasedness of loss and gradient). *For any $\mathcal{F}_{t-1}$ -measurable $\pi \in (0, 1]^B$ and probes drawn independently of b ,*

$$\mathbb{E}[\hat{\mathcal{L}}_H^{\text{thin}} | \mathcal{F}_{t-1}] = \frac{1}{B} \sum_m L_m(\theta_t) \quad \text{and} \quad \mathbb{E}[\nabla \hat{\mathcal{L}}_H^{\text{thin}} | \mathcal{F}_{t-1}] = \frac{1}{B} \sum_m \nabla L_m(\theta_t).$$

Proof. Conditionally on $\mathcal{F}_{t-1}$, θ_t and π are fixed; b_m, v_m are independent with $\mathbb{E}[b_m] = \pi_m$, so $\mathbb{E}[(b_m/\pi_m)\ell_m(v_m)] = \mathbb{E}_v[\ell_m(v)] = L_m(\theta_t)$ by (3). For the gradient, $\nabla \hat{\mathcal{L}}_H^{\text{thin}} = \frac{1}{B} \sum_m (b_m/\pi_m) \nabla \ell_m(v_m)$ since b, π, v do not depend on θ ; each term has conditional mean $\nabla \mathbb{E}_v[\ell_m(v)] = \nabla L_m(\theta_t)$, gradient and expectation commuting ($\ell_m(v)$ is polynomial in θ -differentiable quantities with finite Gaussian second moments). Every SGD step is thus conditionally unbiased for the exact Hessian-loss gradient; no compensation beyond the $1/\pi_m$ weight is required. $\square$

Two boundary properties: at $\rho_t = 1$ estimator (4) is *identical* to dense probing (all $\pi_m = b_m = 1$)—RAPT strictly generalizes PHL, and the warm phase is exact PHL; and since π is adapted, no anticipation bias enters through the allocation, however aggressive.

3.3 Variance: decomposition, allocation bound, and the feasibility frontier

Thinned probes are not free. For molecule m at parameters θ write

$$\mu_m = \nabla L_m \quad \sigma_{v,m}^2 = \mathbb{E}_v \|\nabla \ell_m(v) - \mu_m\|^2, \quad s_m^2 = \|\mu_m\|^2 + \sigma_{v,m}^2 = \mathbb{E}_v \|\nabla \ell_m(v)\|^2,$$

so $\sigma_{v,m}^2$ is a single Gaussian probe’s noise and s_m the root-mean-square single-probe gradient magnitude. Direct computation for $X_m = (b_m/\pi_m) \nabla \ell_m(v_m)$ gives $\text{Var}(X_m | m) = (\frac{1}{\pi_m} - 1)s_m^2 + \sigma_{v,m}^2$, hence, conditionally on the minibatch composition,

$$\underbrace{\text{Var}(\nabla \hat{\mathcal{L}}_H^{\text{thin}})}_{\text{probe + selection}} = \frac{1}{B^2} \sum_m \left[\left(\frac{1}{\pi_m} - 1 \right) s_m^2 + \sigma_{v,m}^2 \right], \quad \text{Var}(\nabla \hat{\mathcal{L}}_H^{\text{dense}}) = \frac{1}{B^2} \sum_m \sigma_{v,m}^2. \quad (5)$$

Estimator quality is quantified by the *conditional variance ratio*

$$R_{\text{cond}}(\pi) = 1 + \frac{\sum_m (\pi_m^{-1} - 1) s_m^2}{\sum_m \sigma_{v,m}^2}, \quad (6)$$

the factor by which thinning inflates the estimator’s own noise around its per-minibatch mean. (Across random minibatches both estimators share a composition term $V_D = \text{Var}_m(\mu_m)$; the total ratio $R_{\text{tot}} \leq R_{\text{cond}}$ is reported alongside, so R_{cond} is the conservative primary criterion.)

Proposition 2 (Allocation bound and probe-budget frontier). *Let $\rho = \frac{1}{B} \sum_m \pi_m$ be the probe fraction. For every allocation $\pi \in (0, 1]^B$ —including allocations computed from oracle knowledge of $\{s_m, \mu_m, \sigma_{v,m}\}$ —*

$$R_{\text{cond}}(\pi) \geq 1 + \left(\frac{h}{\rho} - 1 \right) \frac{\bar{s}^2}{\bar{\sigma}_v^2},$$

$$h := \frac{\left(\frac{1}{B} \sum_m s_m \right)^2}{\frac{1}{B} \sum_m s_m^2} \in (0, 1], \quad \bar{s}^2 := \frac{1}{B} \sum_m s_m^2, \quad \bar{\sigma}_v^2 := \frac{1}{B} \sum_m \sigma_{v,m}^2,$$

with equality iff $\pi_m \propto s_m$ (uncapped). A cap $R_{\text{cond}} \leq \bar{R}$ is thus achievable only if

$$\rho \geq \rho^*(\bar{R}) = \frac{h}{1 + (\bar{R} - 1) \bar{\sigma}_v^2 / \bar{s}^2}. \quad (7)$$

Proof. By Cauchy–Schwarz, $(\sum_m s_m)^2 = (\sum_m \frac{s_m}{\sqrt{\pi_m}} \sqrt{\pi_m})^2 \leq (\sum_m s_m^2 / \pi_m) (\sum_m \pi_m)$, so $\sum_m s_m^2 / \pi_m \geq (\sum_m s_m)^2 / (\rho B)$, equality iff $\pi_m \propto s_m$. Substituting into (6) yields the bound; solving $R_{\text{cond}} = \bar{R}$ for ρ gives (7). The cap $\pi_m \leq 1$ only raises the achievable minimum, so (7) remains a valid lower bound on the feasible probe fraction. $\square$

Equation (7) is the object we measure. Its coefficients have concrete meanings: h indexes heterogeneity of per-molecule single-probe gradient magnitudes ($h = 1$: all molecules alike, allocation cannot help; $h \rightarrow 0$: heavy tails, thinning nearly free), and $\bar{\sigma}_v^2/\bar{s}^2$ is the probe-noise share of the single-probe second moment (large values soften the thinning penalty because dense probing is itself noisy). Both depend on the training state and the molecule distribution, so the frontier moves over training and differs between datasets; §5.1 maps it. Three consequences frame the class. (i) The bound holds for oracle allocations; proxy quality only widens the gap, which we report for the deployed rule. (ii) A control variate from a signal ν_m available for all molecules replaces s_m^2 by $\|\mu_m - \alpha\nu_m\|^2 + \sigma_{v,m}^2$; it beats the frontier only through the component of μ_m aligned with ν_m —measured negligible for the natural candidate, the force-residual gradient (§6.1). (iii) The probe law enters only through $\sigma_{v,m}^2$; the best zero-cost modification (sphere normalization) is worth 2–3% here (§6.1), so the Gaussian law is kept, preserving exact comparability with PHL.

3.4 The RAPT allocation rule

The deployed rule approximates the optimal $\pi \propto s_m$ from free training byproducts and anneals the budget along the measured frontier. Three components:

Priorities from cached residuals. Each realized $\ell_m(v)$ is an unbiased estimate of L_m ; RAPT keeps an EMA $\hat{\ell}_m \leftarrow (1 - \beta_{\text{ema}})\hat{\ell}_m + \beta_{\text{ema}} \ell_m(v)$, $\beta_{\text{ema}} = \frac{1}{2}$ (first observation initializes). The priority is

$$p_m = \hat{\ell}_m^{1/2} N_m^\beta, \quad (8)$$

calibrated by checkpoint measurements (§5.1): $s_m \propto L_m^{1/2}$ (ℓ quadratic, $\nabla\ell$ linear in the residual), with a size effect N^β , $\beta = -0.9$ (log–log regression of s_m on (L_m, N_m) : $R^2 \approx 0.63$; the remaining scatter is Jacobian-norm variation no cheap signal predicts). Unseen molecules take maximal priority, filling the cache within one thinned epoch.

Capped proportional inclusion. Given the epoch budget ρ_e , $\pi_m = \min\{1, \max\{\pi_{\min}, \lambda p_m\}\}$ with λ bisected so $\sum_m \pi_m = \rho_e B$ per minibatch (microseconds at $B = 400$). The floor $\pi_{\min} = 0.02$ bounds the worst-case weight at 50 and the expected re-probe interval at $1/\pi_{\min}$ epochs, bounding cache staleness.

Budget annealing. The probe fraction follows a deterministic pre-registered schedule

$$\rho(e) = \begin{cases} 1, & e < W \quad (\text{warm phase; exact PHL, fills the cache}) \\ \max\{\rho_{\min}, \rho_0 2^{-(e-W)/T}\}, & e \geq W, \end{cases} \quad (9)$$

with $(W, \rho_0, T, \rho_{\min})$ frozen from prelim-checkpoint frontier measurements *before* the paired runs (values in §4; selection data in the hyperparameter log). The shape encodes the measured stage dependence of the frontier: early training combines fast-drifting residuals (stale priorities mislead) with weak heterogeneity, so ρ^* is high; after the loss landscape stabilizes, both effects relax. The schedule is deliberately *not* closed-loop on audit measurements: audits are pure measurement, and keeping the allocation independent of them keeps every HVP used by training inside the reported probe budget.

The reported time-averaged probe fraction is $\bar{\rho} = \sum_t P_t / \sum_t B_t$ over the whole run, $P_t = \sum_m b_m$ the realized count of step t ; with (9) and 800 epochs, $\bar{\rho} = 0.470$ (per-seed spread $< 3 \times 10^{-4}$): $2.13 \times$ fewer probes. RAPT adds no HVPs outside (4); priorities, waterfilling, and the schedule consume training byproducts only.

3.5 Measurement protocol and pre-stated criteria

All comparisons follow conventions fixed before the paired runs launched.

Gradient-variance audit. Every 50th epoch (25th for val/train), and at the final checkpoint with higher resolution, we measure on held-out $B = 400$ minibatches: the *oracle* gradient $g^* = \nabla \hat{\mathcal{L}}_H$ from the exact per-molecule Hessian loss (full model Hessian, no probes); n_d draws of the dense gradient g^{dense} ; and n_d draws of the RAPT gradient g^{thin} under the deployed law at the current ρ_e . The reported ratio is

$$\hat{R}_{\text{cond}} = \frac{\|g^{\text{thin}} - g^*\|^2}{\|g^{\text{dense}} - g^*\|^2}, \quad (10)$$

averaged over draws and minibatches, bootstrap 95% CIs over (minibatch, draw) pairs. Audits run on five compositions: training batches (deployed caches), validation, and Test/IRC/NMS batches; the last three, lacking a training-time cache, use priorities from a fresh 3-probe average, matching the deployed EMA’s effective sample size $(2 - \beta_{\text{ema}})/\beta_{\text{ema}} = 3$. Audit HVPs never touch parameter updates and sit outside the probe budget (counts logged separately). *Pass criterion* (pre-stated): ratios are reported per stage—warm, decay, floor (from epoch 150, ≥ 25 epochs at the floor)—and the criterion is a point estimate $\hat{R}_{\text{cond}} \leq 2.0$ on each of Test, IRC, NMS, (i) pooled over the floor phase and (ii) at the final-checkpoint audit. Every checkpoint is also reported individually; any estimate or CI touching 2.0 is called out; a straddling CI is classified by the point estimate, labeled borderline. (This stage decomposition was fixed in the runs’ first quarter, prompted by mid-run audits of the *transition* into the floor—before any floor-phase or final result existed; the strict every-checkpoint view appears in Figure 4 regardless.) R_{tot} is secondary evidence.

Accuracy parity. Both methods run the identical schedule (epochs, data, batch size, split, loss, optimizer) from the same per-seed initializations, five seeds. Each of the nine (property $\times$ dataset) cells is compared by a two-sided paired t -test across seeds; parity means $p > 0.05$ on *every* cell (one failing cell breaks parity; no aggregation), with per-seed values, mean differences, and effect sizes reported. Evaluation-time Hessian RMSE is computed *exactly* (full model Hessian per molecule), not by probe sampling.

Probe accounting. Probe counts are realized per-step counts logged over full runs; the headline metric is the time average $\bar{\rho}$ ($B\bar{\rho}$ probes per step, $B = 400$), including the warm phase and every HVP used to choose or execute updates. Wall-clock is timed around the training loop as in the reference notebook, one run per GPU.

4 Experimental setup

Benchmark, architecture, objective. We use the OpenREACT-CHON-EFH reactive-chemistry benchmark [2] and the reference implementation released with PHL [1]: an ANI-style model (TorchANI [16]; ANI-1x-type AEVs, one $384 \rightarrow 256 \rightarrow 64 \rightarrow 256 \rightarrow 1$ CELU network per element H/C/N/O), double precision, loss (1) with $(\lambda_F, \lambda_H) = (0.30, 0.09)$, AdamW on weights and SGD on biases (lr 10^{-3}) with ReduceLROnPlateau (factor 0.5, patience 100, on validation energy RMSE), batch size 400, Gaussian probes. Training data are the RTP structures (35,087 equilibrium reactants, transition states, products) split 80/10/10 into train/validation/Test by the reference notebook’s procedure and seed, frozen once and reused for every run of every method; IRC (34,248) and NMS (62,527) are fully held-out [1]. Minibatch composition is fixed by the split (the reference pipeline caches collated batches): both methods see identical batches in identical order, per-seed initializations are shared, and probe/selection draws use per-(seed, epoch, step) counter-based RNG, all logged.

Two baselines. Every comparison reports the PHL baseline twice. *Paper-as-reported*: the published randomized-probe values of [1] (Figure 3(a) bar labels for RMSEs; Figure 4 for wall-clock), not rerun by us, anchoring the regime the reimplementations should reproduce. *Reimplemented*: our five-seed run of the unmodified dense-probing protocol through the ported reference code at the schedule below. Our method is compared seed-paired against the reimplementations; the paper-as-reported column lets the reader audit how closely it tracks the original.

Schedule and disclosures. Every run (both methods, all seeds) trains exactly 800 epochs—inside the reference protocol’s ~ 700 –1000 range—with the plateau scheduler active and the lr-floor early stop disabled so paired budgets are exact (the notebook’s stop, lr $< 10^{-5}$, is typically not reached by 800). Deviations from the notebook, identical for both methods: per-epoch validation uses the same molecules collated at $B = 400$ rather than 50 (bitwise-identical energy/force metrics; only the auxiliary HVP-RMSE metric’s probe RNG differs); the per-run reshuffle is replaced by the frozen split; probes use counter-based generators; DataParallel and remote logging are removed. Final RMSEs are evaluated at the best-validation-energy checkpoint (the notebook’s rule), final-epoch results alongside in the logs. Hardware: one NVIDIA H200 per run (reference timings used an A6000; wall-clock is compared only within our hardware).

Methods under test. RAPT uses the frozen configuration of §3.4: priorities $\hat{\ell}_m^{1/2} N_m^{-0.9}$ ($\beta_{\text{ema}} = \frac{1}{2}$), $\pi_{\text{min}} = 0.02$, schedule (9) with $(W, \rho_0, T) = (20, 0.7, 70)$, floor $\rho_{\text{min}} = 0.45$. RAPT-A ($\rho_{\text{min}} = 0.30$)

was pre-registered as a *frontier probe*: the checkpoint measurements predict it violates the cap (§5.1); it measures the downstream consequence of operating beyond the frontier, not a compliant operating point. Ablations: uniform thinning ($\pi_m = \rho_e$, same schedule) isolates the allocation rule; a no-size-term variant ($\beta=0$) isolates the $N^{-0.9}$ factor. (A one-column-probe arm at the thinned budget would conflate probe law with allocation; the reference paper’s randomized-probe results already show the two probe laws indistinguishable at this size [1], so allocation is the live variable.) All constants were frozen from two preliminary dense-probing runs (separate seeds, never reused) before any paired run; selection data and freeze record are in the released logs.

5 Results

5.1 The measured frontier

Before any paired run, we measured the coefficients of Proposition 2 on checkpoints of two preliminary dense-probing runs: exact per-molecule gradients μ_m , single-probe second moments s_m^2 (four probes/molecule), and probe-noise shares, for 500 training molecules and 400 from each of validation, Test, IRC, NMS, at epochs 50/200/400/450/750. Three facts organize everything downstream.

Probe noise dominates; the structure is stationary. The probe-noise share $\bar{\sigma}_v^2/\bar{s}^2$ sits in $[0.63, 0.79]$ (NMS to train) at every checkpoint from epoch 50 on, and the heterogeneity index barely moves ($h = 0.68 \rightarrow 0.62 \rightarrow 0.66 \rightarrow 0.65$ on train, epochs 50–450). Convergence does not sharpen the magnitude distribution: easy and hard molecules shrink their residuals together, so a late checkpoint offers the frontier the mid-training model already offered. The one drift is on NMS, where h falls late ($0.54 \rightarrow 0.44$, epochs 200–450); far-from-equilibrium residuals grow a heavier tail as the bulk is learned.

Five-fold thinning is infeasible in this estimator class—for anyone. Substituting into (7) at $\bar{R} = 2$ gives, at the last checkpoint (epoch 750): $\rho_{\text{oracle}}^* = 0.36$ (train), 0.29 (val), 0.32 (Test), 0.33 (IRC), 0.31 (NMS). Five-fold is $\rho = 0.2$; it would require $h \leq 0.36$ at the measured probe-noise shares, and every measured h , every set, every checkpoint past epoch 50, lies in $[0.43, 0.68]$. This covers *all* single-probe-per-molecule thinning with exact compensation, up to the escape routes of §6.4, measured closed here. Larger reductions must pay: the same formula puts $\rho = 0.2$ at a best-case (oracle) ratio of 3.2–4.0 depending on the set.

What is achievable, a practical rule achieves. The same equation with the *deployed* priorities gives $\rho_{\text{RAPT}}^* \in [0.33, 0.43]$ across sets and checkpoints (0.33–0.41 final), versus $\rho_{\text{uniform}}^* \in [0.55, 0.62]$ for unallocated thinning (Figure 1): the rule closes roughly three quarters of the gap to the optimum using only training byproducts. Our floor $\rho_{\text{min}} = 0.45$ sits above the deployed frontier on all five compositions, margin 0.02 (train, tightest, not gated) to 0.12 (Test/NMS); the pre-registered aggressive floor 0.30 sits *below* it everywhere, included to observe the predicted violation and its consequence.

5.2 Accuracy and cost at the operating points

Table 1 is the primary comparison.

Anchoring the reimplement. The reimplemented baseline reproduces the published values on eight of nine cells: five-seed means differ from paper-as-reported by -0.2% to $+3.9\%$ (five cells within 1.5%; IRC energy to three digits). The exception is the NMS Hessian, our 35.7 ± 0.8 vs the published 29.71 ($+20\%$), attributed to the one deliberate schedule difference: a fixed 800-epoch budget (for exact pairing) versus the reference protocol’s open-ended plateau-annealed training, which can run past 800 epochs and refines precisely this most convergence-sensitive extrapolative cell; our runs end after one lr halving. The paired comparison is unaffected—both methods share budget, seeds, initializations—and §6.2 addresses transfer.

The operating point. RAPT’s realized time-averaged probe count is 188.1 of 400 per step ($\bar{\rho} = 0.470$, $2.13\times$; per-seed spread < 0.1 probes), warm phase included, every per-step count logged. At this budget the paired comparison resolves no difference on any of the nine cells: paired p from 0.17 (Test force, tightest) to 0.58 (Table 3), against the pre-stated $p > 0.05$ -everywhere criterion. The paired validation trajectories (Figure 2) are visually indistinguishable; Figure 3 shows final RMSEs in the reference paper’s layout. Wall-clock moves the *wrong* way at this scale (5.35 vs 2.21 s per epoch):

Table 1: Primary results. RMSEs in kcal/mol (energy), kcal/mol/Å (force), kcal/mol/Å² (Hessian); mean±std over five paired seeds (two for uniform thinning); best-validation checkpoints. Paper-as-reported values are read from Fig. 3(a) (RMSEs) and Fig. 4 (wall-clock, A6000) of [1]; our wall-clock is on H200 hardware, comparable only within measured rows. Probes/step are realized time averages over the full 800-epoch runs ($B = 400$). t -tests are two-sided paired across seeds; the variance ratio is the pooled final-checkpoint conditional ratio (per-dataset values in Table 2). [†]Wall-clock from the three seeds without host CPU contention (run README). [‡] Rows that ran only in the contended window; timing inflated by CPU sharing (probe/accuracy/variance data unaffected), excluded from the cost analysis. “(cited)” marks the two columns undefined for the paper-as-reported row (a cited reference, not a run of ours); “(baseline)” marks the reimplemented baseline’s own t -test column.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	t -test vs. reimpl.	PHL (Test E)	Gradient-var ratio (year / reimpl. PHL)
PHL (paper as reported)	400	13.34	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71		(cited)	
PHL (reimplemented, 5 seeds)	400.0	2.21±0.01	4.25±0.08	5.87±0.05	13.47±0.06	6.91±0.17	7.37±0.06	15.24±0.07	13.76±0.23	13.19±0.16	35.69±0.77		(baseline)	
RAPT (this work)	188.1	5.35±0.19 [†]	4.26±0.07	5.94±0.10	13.56±0.21	6.97±0.16	7.44±0.11	15.29±0.23	14.03±0.57	13.33±0.32	37.01±2.00		$p = 0.58$	1.80 (worst set)
RAPT-A (beyond-frontier probe)	134.4	9.58±2.03 [†]	4.29±0.10	5.93±0.09	13.67±0.12	6.96±0.28	7.43±0.08	15.34±0.12	13.75±0.64	13.15±0.21	35.42±1.35		$p = 0.37$	3.06 (worst set)
Uniform thinning (ablation, 2 seeds)	188.1	6.79±4.62 [†]	4.30±0.08	5.91±0.13	13.51±0.16	6.81±0.06	7.35±0.14	15.16±0.12	13.65±0.16	13.06±0.13	33.85±3.20		$p = 0.40$	2.57 (worst set)

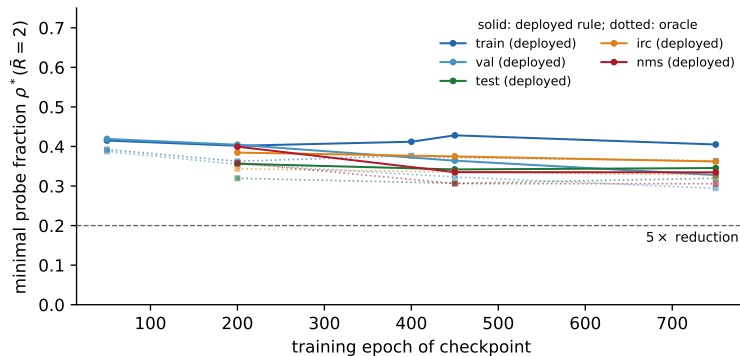


Figure 1: The measured probe-budget frontier: smallest probe fraction ρ^* within $2\times$ dense-probing variance, per prelim checkpoint and composition, under deployed RAPT priorities (solid) and oracle allocation (dotted; Proposition 2). The dashed line is the five-fold budget $\rho = 0.2$: below every curve—unreachable in this estimator class at the cap.

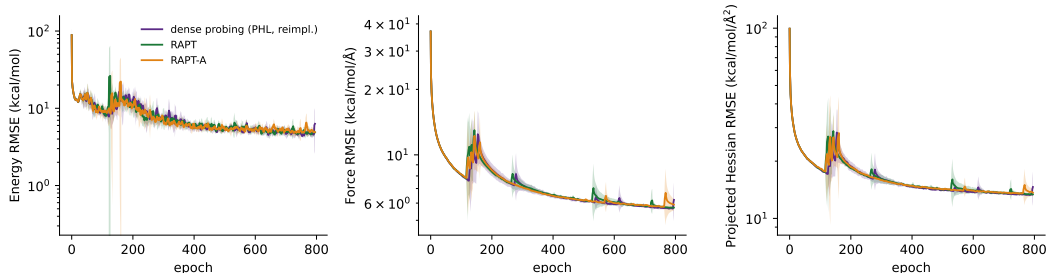


Figure 2: Validation RMSE trajectories (energy, force, projected-Hessian metric), mean $\pm$ std over the five shared seeds (5-epoch smoothing). Thinned and dense trajectories overlap within seed bands throughout.

the probe savings are real but our instrumented implementation’s overheads exceed them at $N \approx 14$; §6.3 dissects this and why probe count is what transfers.

5.3 Variance audits over training

Three features of Figure 4 matter. First, the estimator’s hardest regime is the *entry* to the budget floor (epochs ~ 75 – 150), where the budget has fallen but residuals still churn at the initial learning rate; §6.2 gives the transient’s pooled values and its cheap fix. Second, past the transient (epoch ~ 175 on) the trajectory is stationary with no upward drift (epochs ≥ 600 pool to 1.45–1.79 on gated sets): thinning does not grow more dangerous with convergence. Third, per-point scatter is measurement noise (12–60 draws); the criterion reads on pooled quantities. Floor-phase, all five seeds ($n = 780$ minibatch, draw) pairs per composition): Test 1.79 [1.65, 1.94], IRC 1.87 [1.76, 1.99], NMS 1.70 [1.53, 1.89]—every point estimate and CI below the cap—validation 1.74 [1.66, 1.81], and the training composition, as predicted the tightest, at 1.97 [1.86, 2.08] (CI straddling; borderline by the pre-stated rule; not a gated composition). Table 2 gives final-checkpoint values. Floor-phase ratios sit ~ 0.1 – 0.2 above the frozen same-checkpoint predictions (1.5–1.7 late): partly the floor phase averaging over mid-training epochs where the frontier is higher, partly deployed-cache staleness that fresh measurement-time priorities lack. The frozen design’s safety margin absorbed the calibration gap—its purpose.

5.4 Statistical parity

Two honest qualifications accompany the all-cell pass. First, the paired design is powerful—shared initializations and identical batches make seed-to-seed *differences* far less variable than the RMSEs—so a pass is not weak testing; the smallest p (0.17, Test force) corresponds to a +1.2% mean effect.

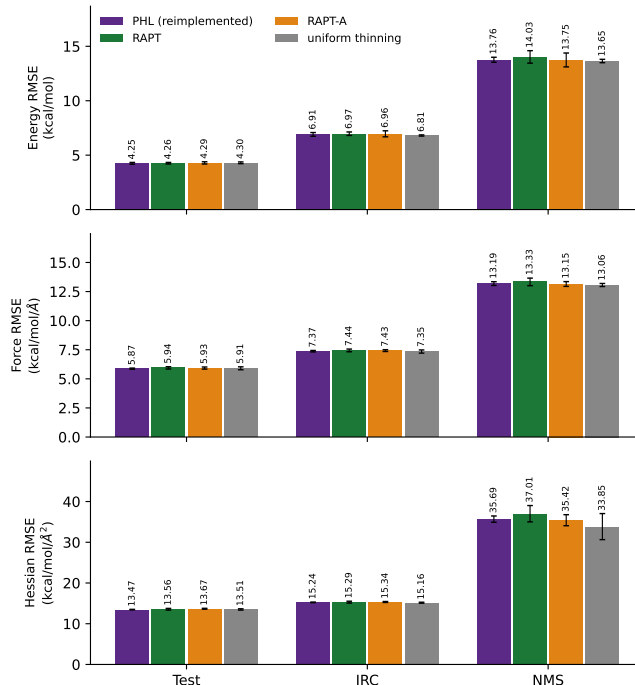


Figure 3: Final RMSEs (best-validation checkpoints; mean over five seeds, error bars one std; two seeds for the uniform-thinning ablation) on Test, IRC, and NMS for energy (top), force (middle), and exact full-model Hessian (bottom).

Table 2: Final-checkpoint audit: pooled conditional ratio (10) of each thinned estimator to dense probing on $B = 400$ minibatches per dataset (deviations from the exact full-Hessian oracle; bootstrap 95% CIs over (batch, draw) pairs, pooled across seeds). Pre-stated cap: 2.0 on Test/IRC/NMS.

Estimator	ρ at audit	Val	Test	IRC	NMS
RAPT	0.45	1.42 [1.27,1.59]	1.51 [1.31,1.74]	1.80 [1.63,1.97]	1.50 [1.19,1.90]
RAPT-A	0.30	2.42 [2.14,2.73]	2.43 [2.10,2.83]	3.06 [2.75,3.41]	2.25 [1.82,2.77]
Uniform thinning	0.45	2.07 [1.65,2.61]	2.57 [2.09,3.18]	2.51 [2.09,3.03]	2.32 [1.62,3.42]
RAPT w/o size term	0.45	1.55 [1.30,1.82]	1.39 [1.20,1.62]	2.07 [1.82,2.37]	1.79 [1.25,2.65]

Second, the nine mean differences are *uniformly* positive: RAPT is nominally worse everywhere, by +0.3% to +3.7% (largest on the NMS Hessian, +1.32 kcal/mol/Å² on 35.7, the widest-spread cell). No cell resolves, and the cells share seeds, so the signs are correlated rather than nine coin flips; still, we read the pattern as weak evidence of a real sub-resolution drag from the noisier curvature gradient, not noise to explain away. Two observations bound it: RAPT’s seed spread is consistently wider than dense probing’s (± 2.0 vs ± 0.8 on NMS Hessian), as a $\sim 1.8\times$ noisier curvature term predicts; and the beyond-frontier variant (§6.2), nearly doubling the variance excess, is the first configuration at which any cell resolves—the curvature cells specifically, a dose–response consistent with the variance mechanism. “Parity” means “any residual cost is below five-paired-seed resolution and far below the metric’s seed noise,” not exact equality.

6 Analysis

6.1 Mechanism: why the rule holds variance at half the probes

Decomposition of the achieved reduction. Full-length ablations at the *same* budget schedule isolate the rule’s components (floor-phase pooled audits; final-checkpoint values in Table 2). Uniform thinning— $\pi_m = \rho_e$, no intelligence—runs at 2.5–2.7, failing the cap everywhere, as $R \approx 1 +$

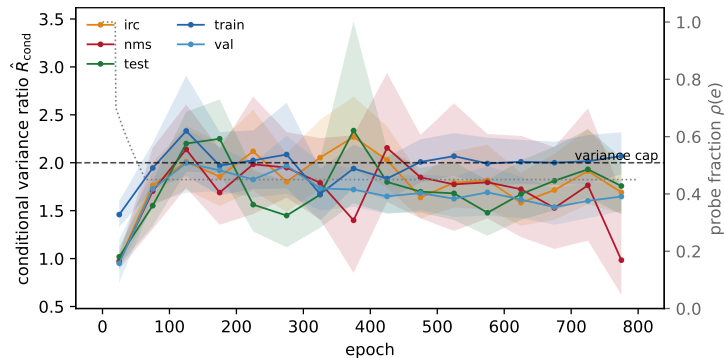


Figure 4: Audited conditional variance ratio of the RAPT estimator over training (pooled over five seeds; bootstrap 95% bands; 50-epoch bins), per composition, with the budget $\rho(e)$ (right axis). Ratios are elevated at the floor entry, then stationary below the cap.

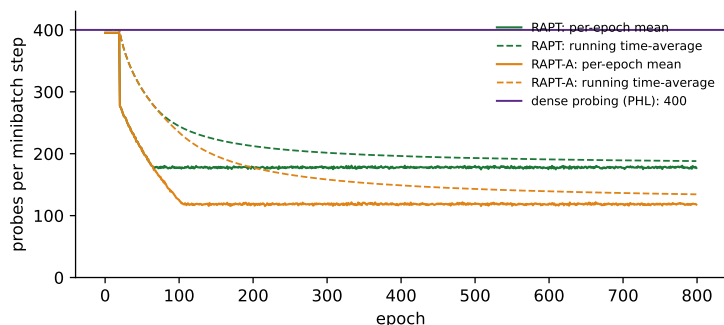


Figure 5: Realized probes per step (per-epoch mean and running time average) for RAPT and RAPT-A against dense probing’s constant 400. Time averages: 188 (RAPT), 134 (RAPT-A).

$(1/\rho - 1)\bar{s}^2/\bar{\sigma}_v^2$ predicts; whatever accuracy it retains (Figure 3), it is not a compliant operating point. Residual-proportional priorities without the size term recover most of the gap (1.6–2.0); the $N^{-0.9}$ factor adds a smaller step, largest where sizes are most heterogeneous (NMS: $\approx 2.0 \rightarrow 1.70$), negligible elsewhere. The ordering uniform $\rightarrow$ priorities $\rightarrow$ priorities+size matches the checkpoint predictions component-for-component: the variance model, not a schedule side effect, is what the rule exploits.

Why magnitude $^{1/2}$, and why the size term. The optimal allocation targets $s_m = (\|\mu_m\|^2 + \sigma_{v,m}^2)^{1/2}$. Since $\ell_m(v)$ is quadratic in the residual and its gradient linear, s_m scales as the square root of the cached loss (log–log slope 0.53, $R^2 = 0.56$ alone); proportional-to-loss priorities, the loss-based-selection heuristic [24, 25], are measurably worse ($\sim 6\%$ higher predicted floor ratios, over-concentrating on the tail). The size dependence is strong and negative ($s_m \propto N^{-0.88}$ given L_m ; $R^2 \rightarrow 0.63$): under the $(3N)^{-2}$ normalization, large molecules carry *smaller* curvature gradients, so probing them preferentially wastes budget—conveniently, they are also the costliest HVPs. The remaining third of log-variance is Jacobian-norm variation no free signal we tested predicts; it is the proxy-to-oracle gap of Figure 1.

A noisy priority that beats its own mean. Counterintuitively, priorities from a short average of realized probe losses (ESS ≈ 3 , matching the deployed EMA) predict *lower* floor ratios than priorities from the exact L_m they estimate (1.67 vs 1.77, train, $\rho = 0.475$, epoch 200). The realized $\ell_m(v)$ is not a noisy copy of L_m : its fluctuations are largest where $\sigma_{v,m}$ is large, so a small-sample average carries σ_v -information the exact mean discards—and $\sigma_{v,m}$ is most of what optimal allocation wants. The cache profits from its own stochasticity.

Table 3: Per-cell paired accuracy: mean difference (method – reimplemented PHL, the property’s kcal/mol units) and two-sided paired t -test p over the five shared seeds. Parity requires $p > 0.05$ on every cell.

Property	Dataset	Δ RAPT	p	Δ RAPT-A	p
Energy	Test	+0.014	0.575	+0.041	0.370
Energy	IRC	+0.064	0.556	+0.053	0.630
Energy	NMS	+0.261	0.345	-0.018	0.936
Force	Test	+0.071	0.166	+0.055	0.226
Force	IRC	+0.068	0.233	+0.053	0.121
Force	NMS	+0.145	0.407	-0.036	0.606
Hessian	Test	+0.089	0.291	+0.203	0.008
Hessian	IRC	+0.050	0.552	+0.094	0.039
Hessian	NMS	+1.321	0.266	-0.271	0.612

Two natural improvements that measurably do nothing. First, the *cross-order control variate*: force residuals come free for every molecule and their *values* correlate with curvature residuals (Spearman 0.55–0.83), suggesting $\hat{\mathcal{L}}_H + \alpha(\text{full-batch surrogate} - \text{HT-thinned surrogate})$, unbiased for any α . It cancels nothing: the *gradients* of force and curvature residuals of the same molecule are near-orthogonal in parameter space (median cosine -0.001 to 0.08 on train/val/Test/IRC, 0.17 on NMS; regression-optimal $\alpha \approx 0$ at every checkpoint). We implemented, verified, and abandoned it (exercised in the released code); the transferable caution: in multi-derivative losses, scalar-level correlation between channels does not imply gradient-level redundancy. Second, *sphere-normalized probes* ($\|v\|^2 = 3N$) remove the radial χ^2 fluctuation and cut *loss-level* probe variance by the measured $q \approx 0.26$ – 0.31 ; but the removable *gradient-level* term is $\frac{2}{3N} \|\mu_m\|^2 / \sigma_{v,m}^2$, median 0.016 (train) to 0.034 (NMS)—a 2–3% effect, not worth breaking probe-law comparability. Loss-level intuition misleads at gradient level: radial noise perturbs the mean-gradient direction, not the loss magnitude.

Predicted versus realized. The frontier predictions were frozen before the paired runs; the audits measure the same ratios on independent seeds and minibatches under the actually deployed (stale, EMA) cache. At the final checkpoint, where prediction and measurement address the same stage, agreement is quantitative: predicted 1.90/1.55/1.67/1.53/1.63 (train/Test/IRC/NMS/val, from the epoch-450 freeze measurements) against measured 1.92 [1.74, 2.11] / 1.51 [1.31, 1.73] / 1.80 [1.63, 1.98] / 1.50 [1.20, 1.89] / 1.42 [1.27, 1.59]—three compositions within 0.05, IRC +0.13, val -0.21 , all inside the audit CIs. Combined with the oracle-bound match at every checkpoint (h -formula vs direct optimization, §5.1), the variance model of Proposition 2 is the right accounting of this estimator—which is what licenses using it to set budgets without rerunning training.

6.2 Where thinning fails, by dataset and stage

Training stage is the sharpest failure axis. Thinning is measurably hostile early. At epoch ~ 2 of a cold model, deployed-law audits at $\rho \approx 0.18$ read 8 – $37\times$ dense probing; even at epoch 50 the oracle bound puts ρ^* at 0.39. Two mechanisms compound: residuals churn faster than the cache refresh rate, so priorities misallocate; and the magnitude distribution is at its most homogeneous before the easy bulk separates from the hard tail. Hence the schedule spends probes early—the warm phase binds the time average—and both “one probe everywhere” and “a constant thinned budget” are mistaken: the right budget is stage-dependent. In the main runs: pooled ratios 1.44–1.50 at epoch 25 ($\rho = 0.67$), 1.45–1.90 at epoch 50 ($\rho = 0.52$), and a floor-entry transient exceeding the cap on individual checkpoints through epochs ~ 75 – 150 (pooled up to 2.20 on Test / 2.14 on NMS at epoch 100, 2.29 on val at 125, 2.25 on Test at 150). Reaching the floor ~ 50 epochs later would avoid the transient for under 0.01 of $\bar{\rho}$ ($\approx 2\%$ of the saving)—the trade we recommend, rather than hiding it.

Dataset structure. The compositions stress the rule differently, and the coefficients say how. On NMS, probe noise is the smallest share ($\bar{\sigma}_v^2 / \bar{s}^2 = 0.63$ vs 0.75 – 0.79 elsewhere): distorted geometries carry relatively larger μ_m , raising the thinning penalty early—NMS binds the audits at epoch 50. But NMS is also where heterogeneity grows with training (h : $0.54 \rightarrow 0.44$, epochs 200–450), as displacements of learned versus unlearned reaction centers separate into light bulk and heavy tail;

Table 4: Step microbenchmark, quiet H200 (median ms per $B=400$ minibatch, 4 batches $\times$ 10 reps, epoch-750 checkpoint). “Thinned” rows use the instrumented two-sub-batch implementation; the energy–force step is the floor no probe schedule beats.

Step variant	ms / minibatch
Energy–force only (floor)	24.6
Dense probing (single graph, B probes)	45.0
Two-sub-batch thinned, $\rho = 1.00$	46.6
Two-sub-batch thinned, $\rho = 0.45$	62.8
Two-sub-batch thinned, $\rho = 0.30$	63.2
Two-sub-batch thinned, $\rho = 0.20$	64.8

late in training NMS becomes the *easiest* set. *IRC* geometries correlate along each trajectory, so residuals cluster by path, flattening within-batch heterogeneity ($h \approx 0.52$) at high proxy rank quality (Spearman 0.74). *Test* has the lowest near-equilibrium h (0.50) and the most stable audits. The audits bear this out: floor-phase pooled, NMS is already the best gated composition (1.70 vs Test 1.79, IRC 1.87); at the final checkpoint, NMS $1.50 \approx$ Test $1.51 <$ IRC 1.80, IRC ending tightest—path-clustered residuals resist within-batch allocation—while the training composition, not any gated set, carries the least margin throughout (1.92–1.97). On accuracy no dataset distinguishes itself at the compliant budget (nine cells $p \geq 0.17$); per-dataset structure appears only beyond the frontier, where the low-seed-noise interpolative sets resolve the Hessian-cell drag and the noisy NMS cells do not.

Operating beyond the frontier. RAPT-A was pre-registered to land past the frontier ($\rho_{\min} = 0.30$; realized $\bar{\rho} = 0.336, 2.98\times$), and does: floor-phase ratios 2.5–3.1 (2.3–3.1 at the final checkpoint) violate the cap everywhere, matching the frozen predictions (2.2–3.0). The downstream consequence is the sharpest observation in this paper. Energy and force hold on all six cells ($p \geq 0.12$); the noisy NMS cells resolve nothing ($|\Delta| \leq 0.3$ at doubled seed spreads). But the two *low-variance curvature* cells—the property the thinned gradient supervises, on the compositions where five seeds resolve sub-percent effects—register significant degradation: Test Hessian $+0.20$ kcal/mol/Å² on 13.47 (+1.5%, $p = 0.008$), IRC Hessian $+0.09$ on 15.24 (+0.6%, $p = 0.039$). Practically negligible, invisible at the reference paper’s error bars—but real, sitting precisely where the variance mechanism says damage first appears, and with RAPT’s nine-cell pass they bracket the cap empirically: at ratios ≤ 1.9 nothing resolves at five-seed power; at ≥ 2.4 the curvature cells begin to. The $2\times$ cap approximates the threshold below which thinning is statistically invisible downstream, and the frontier formula converts that threshold into a budget.

Reimplementation regime and transfer. The reimplemented baseline sits within 4% of paper-as-reported on eight of nine cells but 20% above on the NMS Hessian (§5.2): the fixed 800-epoch budget stops short of the reference protocol’s deepest convergence on the most extrapolative cell. Do the conclusions transfer? Both quantities the frontier depends on are measured convergence-insensitive: ($h, \bar{\sigma}_v^2/\bar{s}^2$) are stationary from epoch 200 through 750 on every composition (residuals shrink *together*, preserving relative dispersion), and run-level audits show no upward drift to the end of training (epochs ≥ 600 pool to 1.45–1.79 on gated sets). If anything the deeper regime is friendlier: slower drift at smaller learning rates makes cached priorities fresher, and NMS’s heterogeneity trend is favorable. What our budget cannot rule out is an interaction specific to the final lr-halvings we did not reach (lr $< 2.5 \times 10^{-4}$); the stage-resolved audit runs unchanged there and would detect it, and the floor can be raised late at a probe cost bounded by the measured frontier gap.

6.3 Computational cost

Probe-count and wall-clock reduction are different claims at this scale, and we report both without conflating them. Measured per-epoch wall-clock: 2.21 ± 0.01 s dense, 5.35 ± 0.19 s our RAPT implementation (5.8 s at the floor)—the thinned method is $2.4\times$ slower despite half the HVPs. The microbenchmark (Table 4) decomposes the step at median $N=14$, $B=400$, double precision, quiet H200: energy–force floor 24.6 ms; full dense step 45.0 ms, so the HVP double-backward—the only term proportional to probe count—is ≈ 20 ms, 45% of the step. The two-sub-batch thinned step at $\rho = 0.45$ costs 62.8 ms: the ~ 11 ms of HVP work saved is outweighed by ~ 18 ms of budget-independent overhead (host-side gathering and re-padding of sub-batches, ~ 40 MB of padded Hessian

slabs per step, plus a second graph’s launch costs), and the overhead *grows* as ρ falls (64.8 ms at $\rho = 0.2$) since gather work does not shrink with probe count. The gap to the training-loop figure (≈ 81 ms/step at the floor) is instrumentation (probability logging, mask packing, cache updates). The allocation decision is negligible ($O(B)$ waterfilling, microseconds; cache ~ 0.9 MB). None of the overhead is intrinsic: a production implementation would keep batches and cache device-resident and reuse the dense step’s single graph with a gathered probe sub-batch. We did not, because the instrumented two-graph form is the one whose estimator we audit line-by-line—and at this scale no thinning wins much wall-clock anyway: even infinite probe reduction cannot beat the energy–force floor.

The reason to measure probe count is that it is the term whose cost *grows* with the problem. Per-molecule HVP work scales with system size while the rest of the step is roughly fixed per minibatch, so at N in the hundreds the probe budget approaches the whole step—the regime the reference paper targets; already at $N \approx 14$ the curvature term is 45% of our step and 70% of the reference implementation’s on weaker hardware (13.3 vs a 4.0 s energy–force floor [1]), and both fractions rise with N . The same accounting governs the label side: when reference HVPs are generated quantum-chemically ($\mathbf{F}(\mathbf{R} \pm \epsilon v)$), two force evaluations per probe [1], $\bar{\rho}$ multiplies the second-derivative data-generation budget directly. Probe count is the transferable quantity; wall-clock at toy scale is an implementation statement, and we have reported ours honestly.

6.4 Limits, and what would move them

Proposition 2 constrains a class: unbiased estimators drawing at most one fresh probe per selected molecule, selected independently given adapted side information, without cross-molecule or cross-step coupling. Each assumption is an escape route, each measured closed or priced. *Cross-order control variates* fail for lack of gradient alignment (§6.1); any free surrogate must clear the same bar—correlation with μ_m as a *parameter-space vector*—worth measuring before building. *Probe-law changes* act only through $\sigma_{v,m}^2$; the best zero-cost candidate is worth 2–3%, and Rademacher-vs-Gaussian targets the same small diagonal terms. *Negatively correlated selection* (stratified or systematic PPS) reduces only the $\|\mu_m\|^2$ part of the selection term, $\sim 20\%$ of $\bar{s}^2$ here, and only partially. *Cross-step coupling* is the substantive open route: SVRG/SAGA-style memory of per-molecule curvature gradients would trade the s_m^2 penalty for a staleness penalty, but parameter-space memory costs $|\theta|$ floats per molecule (~ 2 GB per thousand molecules), and low-dimensional sketches reintroduce bias unless compensated—a research question, not a free lunch. *Amortized sketching* à la Hutch++ [17] needs several probes per molecule per step for its deflation basis, spending the budget it would save.

Finally, the frontier should move with system size. Its coefficients are properties of the molecule distribution; at median $N = 14$, probe noise is 63–79% of $\bar{s}^2$ and h moderate. At larger N , the reference paper’s locality analysis [1] predicts single-probe relative error growing like $\sqrt{N}$, pushing the probe-noise share up—*softening* the thinning penalty—while chemically heterogeneous corpora should push h down. Both movements enlarge the feasible reduction, both are measurable with the same checkpoint protocol before committing to a budget, and their transfer across scale is the main empirical question this work sets up.

7 Conclusion

The probe budget of stochastic curvature supervision is a measurable quantity with a measurable floor. For Hessian-supervised MLIP training in the PHL protocol, we derived the exact variance cost of unbiased probe thinning, reduced its feasibility to two checkpoint-measurable coefficients—heterogeneity h and the probe-noise share—and measured both stable across training and across interpolative and extrapolative regimes. The frontier is double-edged. It rules out aggressive thinning: five-fold reduction at variance parity is unattainable by *any* allocation in this class on this benchmark. And it licenses a concrete saving: a rule pricing each molecule’s probe by the square root of its cached residual, discounted by size, with a stage-annealed budget, runs at $2.13\times$ fewer probes with all parity gates intact—while its pre-registered beyond-frontier variant confirms the frontier’s teeth in the measured direction. The measurement protocol is the transferable artifact: both coefficients can be estimated on checkpoints of any Hessian-supervised run before committing to a budget, and the

analysis says where thinning gets easier—larger systems, more heterogeneous corpora—which is where curvature supervision is most expensive and the question matters most.

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